

CSC 402 Lab 2: Kotlin & Jetpack Compose Fundamentals

Student: Narjes Al-Wohaib

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Major: Computer Science

Level: 7

Part A: Kotlin Console Tasks

Task 1 Output

```
--- Task 1: Student Information ---  
Name: Narjes Al-Wohaib  
ID: 2240002170  
GPA: 4.42  
Credits: 78  
Status: Active
```

Task 3 Output

```
--- Task 3: Score Calculation ---  
Student Narjes achieved a weighted score of 91.40
```

Task 4 Output

```
--- Task 4: List of Languages ---  
Original list: [Kotlin, Java, Python, C++, Swift]  
First language: Kotlin
```

```
Last language: Swift  
Sorted list: [C++, Java, Kotlin, Python, Swift]  
Does the list contain Kotlin? Yes
```

Task 5 Output

```
--- Task 5: Course Management ---  
Courses with >= 3 credits: [CSC 402, CSC 311, CSC 340, MATH 202]  
All course titles: [Mobile Application Programming, Database Systems, Operating Systems, Discrete  
Total registered credits: 15  
Search Result (CSC 402): Mobile Application Programming  
Updated Course Schedule: CSC 402 - Mon / Wed / Fri
```

Part B: Jetpack Compose Tasks

Task 6 — Welcome Screen

2:39  

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Task 6 - My First Screen



Hello, Jubail!

Welcome to Mobile Application
Programming

Built with Jetpack Compose

Everything on this screen is drawn
by one composable function.



Task 7 — Student ID Screen

 Task 7 Screenshot

Task 8 — Counter Screen

 Task 8 Screenshot

Task 9 — Greeting Screen

3:32  

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Task 9 - Text Input & State

Welcome, Narjes!

Letters entered: 6

Your Name

Narjes

Clear



ment: The name variable is updated on every
oke, causing the UI to recompose and update the
ng and count instantly.



Task 10 — Course List Screen

 Task 10 Screenshot

Task 11 — Task Tracker Screen

 Task 11 Screenshot

Questions & Answers

Q1: What is the main difference between 'val' and 'var' in Kotlin?

'val' is used to declare immutable (read-only) variables, meaning their value cannot be changed after initial assignment. In contrast, 'var' declares mutable variables which can be reassigned as needed throughout the code.

Q2: Why are data classes useful in Kotlin application development?

Data classes are highly efficient for holding state because the compiler automatically generates standard methods like equals(), hashCode(), toString(), and copy(). This reduces boilerplate code significantly when creating models for UI lists or database entities.

Q3: How does the 'remember' function help maintain UI state during recomposition?

'remember' stores a value in the Composition during the initial call and returns the stored value during subsequent recompositions. This ensures that local state, such as a counter value or text input, is not reset every time the UI refreshes.

Q4: Explain the role of 'mutableStateOf' in Jetpack Compose.

'mutableStateOf' creates an observable state holder that Jetpack Compose tracks for changes. When the value inside the state holder is updated, Compose automatically triggers a recomposition of all UI elements that read that specific value.

Q5: When should you use 'LazyColumn' instead of a standard 'Column'?

'LazyColumn' is preferred for long or infinite lists because it only composes and lays out items that are currently visible on the screen. A standard 'Column' renders all items immediately, which can cause performance issues and memory lag for large datasets.

Q6: What is 'State Hoisting' and why is it recommended?

State hoisting is the pattern of moving state to a component's caller to make the component itself stateless. This is recommended because it makes composables more reusable, easier to test, and allows the parent to manage the source of truth for the entire screen.

Q7: Describe a scenario where you would use a 'derived state' in a Compose screen.

A derived state is useful when one value depends entirely on another, such as calculating a 'Total' sum from a list of items or checking if a text input is valid. In Task 9, the letter count was a derived state that updated automatically as the user typed their name.

Q8: How does 'Modifier.weight(1f)' affect the layout within a Row or Column?

'Modifier.weight(1f)' instructs the layout engine to give the element all remaining available space after other non-weighted siblings have been measured. This is essential for creating flexible UI components, such as a text label that expands to fill the width of a card next to a fixed-size icon.

AI Use & Disclosure Log

I used ChatGPT and Android Studio Gemini to assist with project setup, UI component structuring, and troubleshooting emulator connectivity issues. These tools helped in generating boilerplate for LazyColumn and ensuring Material 3 styling consistency across screens.